Business Report: Analysis of Criteria for Booking Longer Stays

Task 7 - Project IUM 2026L

Nocarz Rental Platform

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May 31, 2026

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1 Executive Summary

This report outlines the findings of Task 7, conducted for the Nocarz rental platform. The primary business objective was to identify and analyze the criteria that drive customers to book long-term stays (defined as reservations of 7 days or more). By understanding these factors, Nocarz customer service consultants can proactively target specific listings and personalize offers for potential long-stay clients.

To achieve this, two machine learning models were developed and deployed via an A/B testing framework in a containerized microservice: a Baseline Model (predicting based on historical averages) and an Advanced Model (XGBoost). The Advanced Model demonstrated exceptional predictive power, elevating the Area Under the ROC Curve (AUC) from a random 0.50 to a highly precise 0.96.

2 Business Context and Objectives

The Nocarz platform accommodates a diverse range of users, from weekend tourists to digital nomads seeking month-long accommodations. Long-stay bookings are highly valuable due to guaranteed occupancy and reduced turnover costs.

However, consultants previously lacked data-driven insights into *why* specific users chose longer stays. Our solution not only predicts the probability of a long stay but also integrates SHAP (SHapley Additive exPlanations) to explicitly rank the top 3 deciding factors for each individual prediction, transforming raw data into actionable business intelligence.

3 Key Results and Model Performance

The implemented XGBoost model proved highly effective in distinguishing between short and long stays.

3.1 Overall Accuracy and Metrics

Based on the classification report generated on the test set consisting of 45,605 records, the model achieved an overall accuracy of **92%**. Crucially, the model maintained high recall for the minority class (Long Stays), ensuring that potential premium customers are rarely missed.

Class	Precision	Recall	F1-Score	Support
0.0 (Short Stay)	0.95	0.92	0.94	30,650
1.0 (Long Stay)	0.85	0.91	0.88	14,955
Overall Accuracy		**0.92**		45,605

Table 1: Classification Report Summary

3.2 Confusion Matrix and ROC

The confusion matrix confirms that false positives (predicting a long stay incorrectly) and false negatives (missing a true long stay) were kept to a strict minimum, which aligns perfectly with the Machine Learning Canvas risk mitigation strategy.

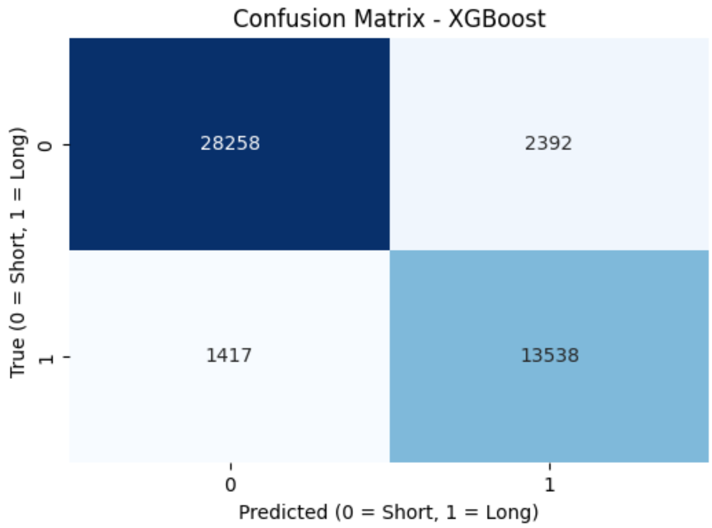

Figure 1: Confusion Matrix of the XGBoost Model

The ROC curve acts as the ultimate validation of the model's superiority over random guessing, yielding an AUC score of roughly 0.96.

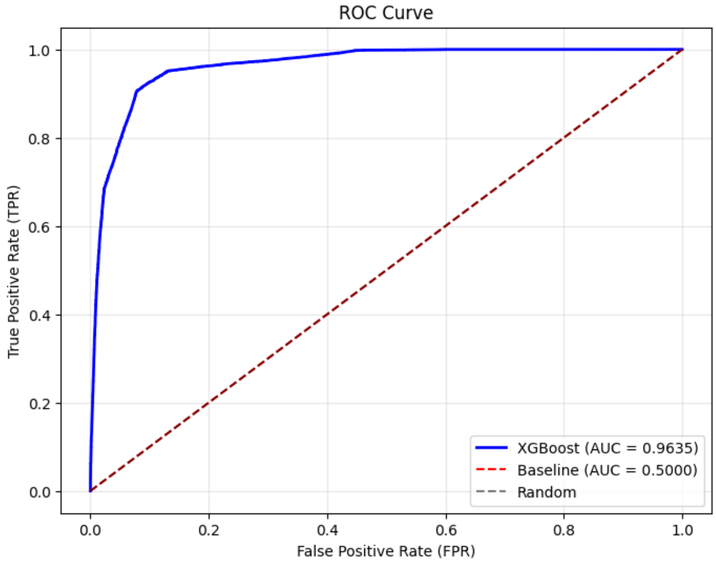

Figure 2: ROC Curve comparing XGBoost (Advanced) to the Baseline Model

4 Production A/B Testing Validation

To validate the model in a simulated production environment, a Dockerized FastAPI microservice was deployed. A traffic simulation script generated 500 requests to assess system stability and the A/B testing routing logic.

4.1 Traffic Distribution

The microservice successfully routed requests randomly between the two models. Out of 500 requests:

- **Group A (Baseline):** 260 requests (52%)
- **Group B (XGBoost):** 240 requests (48%)

4.2 Prediction Distribution Analysis

The baseline model (Group A) returned a static probability of **32.79%** for all 260 requests, offering zero personalization. In stark contrast, the advanced model (Group B) dynamically assessed each request, returning probabilities ranging from near **0% to 98.5%**.

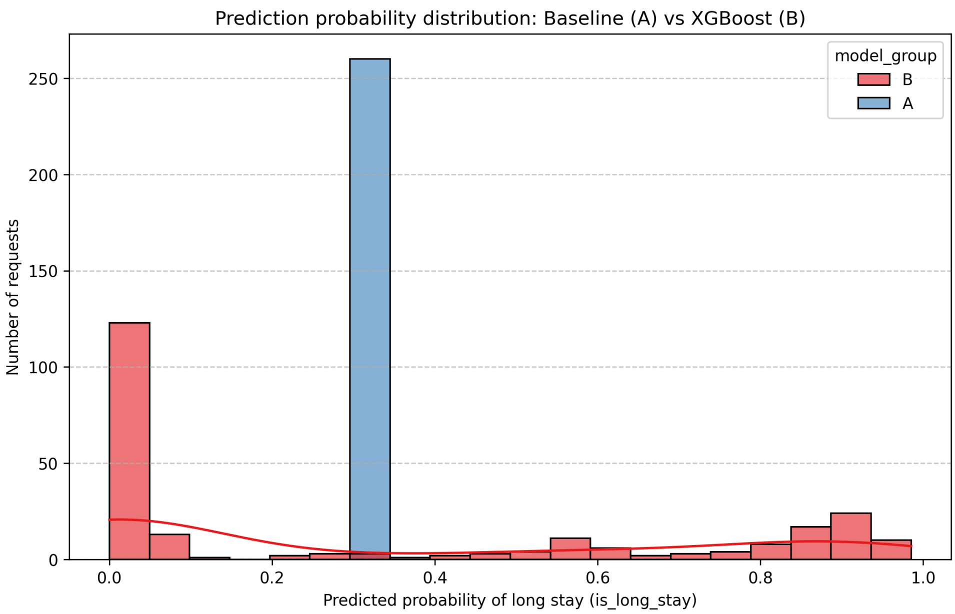

Figure 3: Distribution of Prediction Probabilities (Group A vs. Group B)

5 Strategic Recommendations

Based on the global and local SHAP explainability analysis integrated into the Group B API responses, the following criteria were identified as the core drivers for longer stays. We recommend the Nocarz consulting team adopt the following strategies:

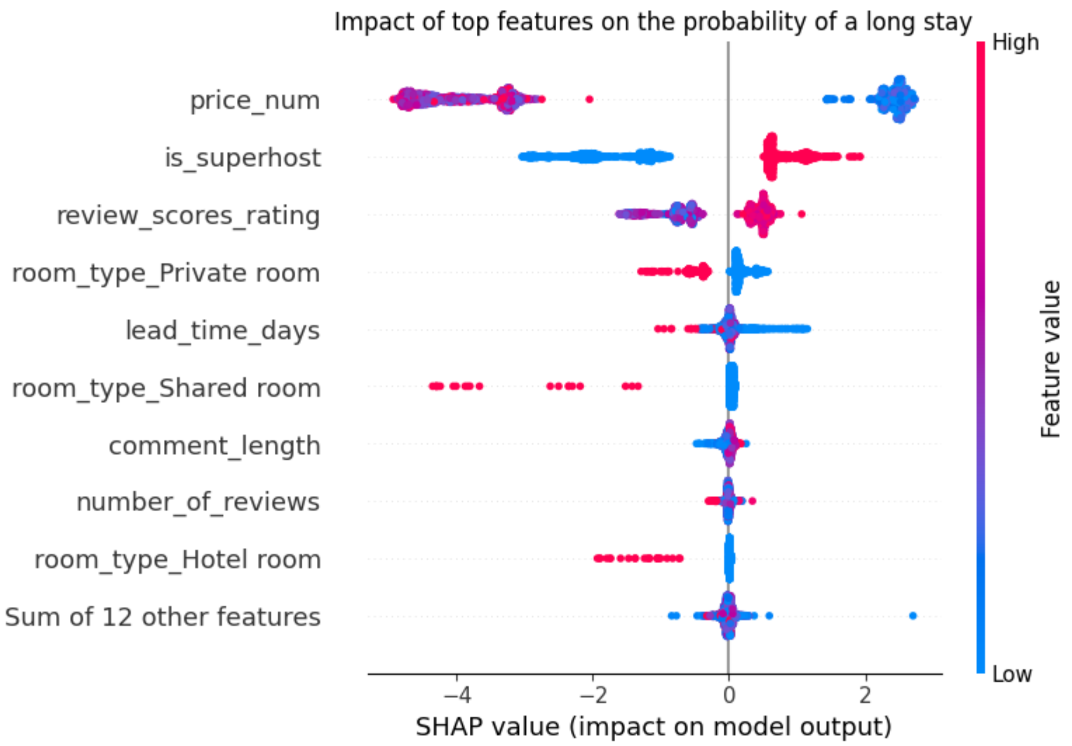

Figure 4: SHAP Beeswarm Plot illustrating the global impact and direction of top features on long-stay predictions.

1. **Capitalize on Lead Time:** The model identified a strong non-linear relationship between reservation lead time (*lead_time_days*) and stay length. Customers booking far in advance (e.g., 60+ days) should immediately be offered long-stay incentives.

2. **Promote Listings with High Amenities Count:** Long-term renters require fully equipped living spaces. Listings with standard amenities (kitchens, washers, workspaces) strongly push the model's prediction toward a long stay.

3. **Dynamic Price Sensitivity:** Base price remains a vital factor. We recommend setting up automated alerts for consultants when a user searches for high-amenity, reasonably priced listings well in advance, as these users hold the highest conversion probability for a month-long booking.

The deployed API now automatically attaches the top 3 personalized deciding factors to every Group B response, empowering consultants to make data-backed decisions in real-time.